

Strain-Sensitive Magnetization Reversal of a van der Waals Magnet

Yu Wang, Cong Wang, Shi-Jun Liang, Zecheng Ma, Kang Xu, Xiaowei Liu, Lili Zhang, Alemayehu S. Admasu, Sang-Wook Cheong, Lizheng Wang, Moyu Chen, Zenglin Liu, Bin Cheng,* Wei Ji,* and Feng Miao*

By virtue of the layered structure, van der Waals (vdW) magnets are sensitive to the lattice deformation controlled by the external strain, providing an ideal platform to explore the one-step magnetization reversal that is still conceptual in conventional magnets due to the limited strain-tuning range of the coercive field. In this study, a uniaxial tensile strain is applied to thin flakes of the vdW magnet Fe_3GeTe_2 (FGT), and a dramatic increase of the coercive field (H_c) by more than 150% with an applied strain of 0.32% is observed. Moreover, the change of the transition temperatures between the different magnetic phases under strain is investigated, and the phase diagram of FGT in the strain–temperature plane is obtained. Comparing the phase diagram with theoretical results, the strain-tunable magnetism is attributed to the sensitive change of magnetic anisotropy energy. Remarkably, strain allows an ultrasensitive magnetization reversal to be achieved, which may promote the development of novel straintronic device applications.

Van der Waals (vdW) magnetic materials, including CrI_3 , $\text{Cr}_2\text{Ge}_2\text{Te}_6$, and Fe_3GeTe_2 , etc., have attracted much attention over the past few years for offering a platform to explore new fundamental physics and novel device applications.^[1–8] Layered

vdW magnets are more amenable to external stimuli.^[9] With this physical property, several means such as electrostatic doping,^[3,10–12] vertical electric field,^[13] hydrostatic pressure,^[14–16] etc., have been applied on vdW magnets to engineer their magnetic properties in an attempt to switch the magnetization direction. Notably, controllable switching of magnetization direction establishes the fundamental of “writing” operations which is paramount in the spintronic technologies.^[7] In this aspect, exploring more strategies for modifying and switching the vdW magnetism is of crucial importance to not only understand fundamental magnetic properties but also develop new types of functionalities in vdW magnets based spintronic devices.^[7,17]

Strain has been demonstrated to be an alternative approach to tuning the

magnetism of conventional magnetic thin films.^[18] It plays an important role in expanding the functionalities in the spintronic device,^[7] since the effective tuning of the easy axis of magnetization via strain in magnetostrictive nanomagnets enables the “straintronic” switching.^[19,20] Compared with the conventional magnets, tuning of physical parameters in the vdW magnets, i.e., magnetic anisotropy energy (MAE), spin-exchange coupling (SEC), could be sensitive to strain-induced lattice deformation.^[21–24] Theories have predicted that the strain-induced change of the SEC allows for observation of transitions between ferromagnetic and antiferromagnetic magnetic configurations in CrI_3 .^[22,23] Moreover, the strong dependence of MAE on strain would lead to a significant change of H_c .^[18] This large tuning range of H_c is the critical for reversing the magnetization in one step,^[11] which is the fundamental of writing operation for data storage in magnetic memory devices. It is noted that such one-step magnetization reversal has been realized by Curie point writing^[25] and electric field,^[26] but is still conceptual for the strain-assisted version,^[27] hindering the development of practical straintronic devices.

Here, we applied tensile strain on vdW magnet Fe_3GeTe_2 (FGT) flakes and investigated the tuning of the magnetic properties herein. By taking the Hall measurement, we found remarkably increased coercive field H_c , Curie temperature T_c , and transition temperature T_1 between single- and labyrinthine-domain states under the tensile strain. Density functional

Y. Wang, Dr. S.-J. Liang, Z. Ma, K. Xu, X. Liu, L. Zhang, L. Wang, M. Chen, Z. Liu, Dr. B. Cheng, Prof. F. Miao
National Laboratory of Solid State Microstructures
School of Physics
Collaborative Innovation Center of Advanced Microstructures
Nanjing University
Nanjing 210093, China
E-mail: bincheng@nju.edu.cn; miao@nju.edu.cn

C. Wang, Prof. W. Ji
Beijing Key Laboratory of Optoelectronic Functional Materials
& Micro-Nano Devices
Department of Physics
Renmin University of China
Beijing 100872, P. R. China
E-mail: wji@ruc.edu.cn

Dr. A. S. Admasu, Prof. S.-W. Cheong
Center for Quantum Materials Synthesis, and Department of Physics
and Astronomy, Rutgers
The State University of New Jersey
Piscataway, NJ 08854, USA

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/adma.202004533>.

DOI: 10.1002/adma.202004533

theory calculations suggested that this strain-tunable magnetism in FGT arises from strain induced change of MAE, which may result from the enhancement of the spin-orbit coupling with the lattice expansion in FGT. Finally, we achieved an ultra-sensitive strain-assisted magnetization reversal, which is inaccessible in conventional magnetic thin films.

In this work, we use FGT thin flakes to experimentally investigate the change of magnetic properties caused by tensile strain. FGT is a vdW ferromagnetic (FM) metal with a layered hexagonal crystal structure, where the Fe_3Ge slabs are sandwiched by Te layers (Figure 1a).^[21,28–33] The Curie temperature T_c is relatively high, ranging from 150 to 220 K depending on the Fe occupancy,^[34,35] and decreases sharply when the sample is thinned to 2D limit.^[3,4] Together with the nearly square-shaped hysteresis loop, large coercivity and strong perpendicular magnetic anisotropy, FGT exhibits a potential for vdW heterostructure based spintronics.^[36] In order to apply tensile strain on FGT, we exfoliated FGT thin flakes (about 50 nm) directly onto the polyimide (PI) membrane, which is suitable for using as flexible substrate due to the excellent thermal stability and mechanical properties.^[37,38] A conventional electron-beam lithography and evaporation processes are adopted to fabricate 5 nm Ti/80 nm Au electrodes, with photo image shown in Figure 1b. Raman measurement and energy-dispersive X-ray

spectroscopy were carried out to guarantee the quality of the FGT flakes (see Figures S1 and S2 in the Supporting Information).

The FGT device on PI substrate was then loaded to a home-made strain setup with a three-points bending geometry, as shown schematically in Figure 1c. By pushing the needle to the center of the substrate, we can bend the PI membrane and stretch its top surface, as shown in Figure S3 (Supporting Information), so that a uniaxial tensile strain could be transferred to the FGT sample which is on that top surface of the PI membrane. Notably, the metal electrodes deposited on the sample were also used as clamping points to rule out the slippage between the sample and substrate.^[39] The strength of uniaxial tension strain caused by flexible membrane bending can be described as $\varepsilon = t/2R$,^[39–42] where t is the thickness of the PI substrate and R is the radius of curvature of the bended substrate. In our setup, R is determined by the movement of the pushing-needle toward the substrate (see Figure S3 in the Supporting Information).

We characterized the magnetic properties of FGT thin flakes by taking the Hall measurement.^[3,36] The Hall resistance R_{xy} is the sum of normal Hall resistance and anomalous Hall resistance which are proportional to the external magnetic field B_z and the spontaneous magnetization M_z , respectively, i.e., $R_{xy} = R_0 B_z + R_s M_z$. It is noted that the normal Hall resistance

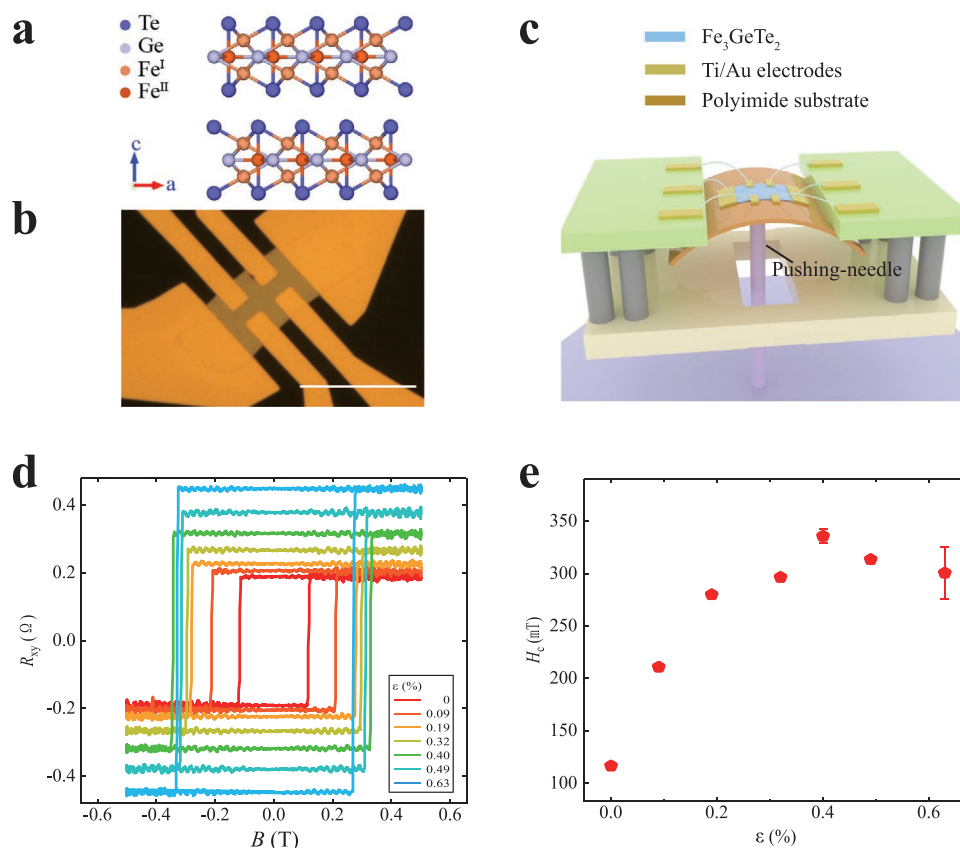


Figure 1. Straining FGT devices. a) Side view of the atomic lattice of FGT. b) Optical microscopy image of a typical FGT device on the polyimide substrate. The scale bar is 10 μm . c) Schematic diagram of device in the strain experimental set-up. d) The R_{xy} versus B curves in the strain regime from 0% to 0.63% at 1.5 K. Strain-induced changes in the hysteresis loop can be observed. e) Coercive force as a function of strain at 1.5 K. The error bar is determined by the left and right boundaries of hysteretic loop. The large error bar with 0.63% arises from the shift of the hysteresis loop (see Note S4 in the Supporting Information for details).

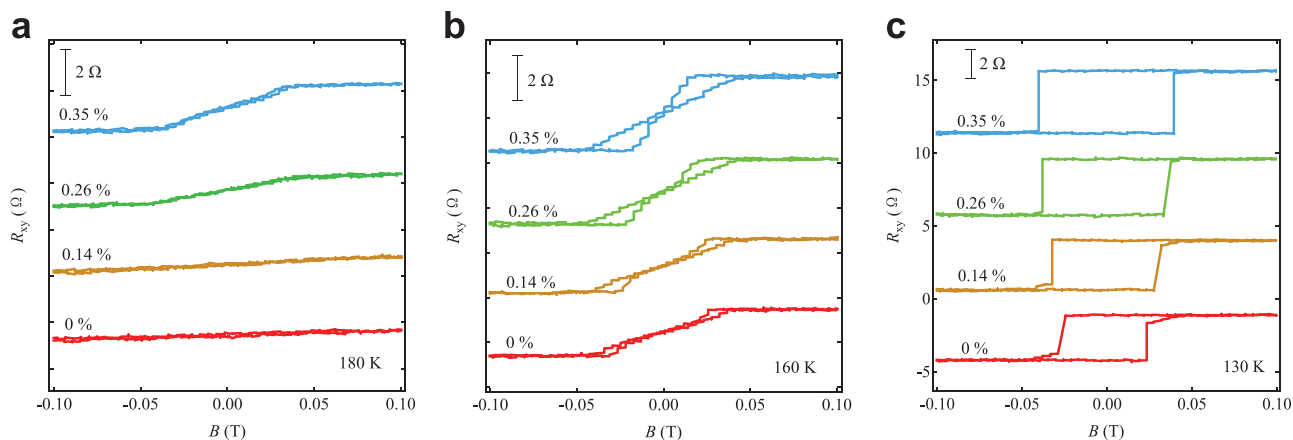


Figure 2. Controlling ferromagnetism in FGT by strain at 180, 160, and 130 K. a) The hysteresis loops with different strains at 180 K. The ferromagnetism of FGT increases gradually with increasing strain. b) The R_{xy} versus B curves in the strain regime from 0% to 0.35% at 160 K. c) Strain dependence of the magnetic hysteresis curves measured by R_{xy} of FGT at 130 K. With the increase of strain (from 0% to 0.35%), the loops gradually tend to rectangular hysteresis loops.

induced by B_z could be negligible compared with the anomalous Hall resistance within the range of magnetic field in FGT,^[36] enabling us to study the spontaneous magnetization of the FGT flake through the Hall measurement. We first examined R_{xy} versus B under different tensile strains at 1.5 K, and the square-shaped hysteresis loops, as shown in Figure 1d, indicated a single magnetic domain^[4] over the entire flake under all the strains. The H_c , which is extracted from the near-vertical jumps in the hysteresis loops, changed from 117 to 300 mT as the strain was increased from zero to 0.32% (Figure 1e). Such enhancement of H_c under tensile strain in FGT is much larger compared with the conventional magnetic materials used in spintronic devices such as CoFe_2O_4 (CFO).^[43]

Investigating the influence of the tensile strain on the magnetic transitions is of critical importance to understand the underneath mechanism of the strain tuning. We set the temperature to 180, 160, and 130 K, respectively, which are around the temperatures of the transitions between paramagnetic, labyrinthine- and single-domain states under zero strain.^[4] As shown in Figure 2a, R_{xy} is approximately linear to the external magnetic field without strain at 180 K, signifying a paramagnetic state. With the tensile strain turned on, the curve of R_{xy} versus B gradually became nonlinear, showing a transition to the ferromagnetic (FM) state. The transition from paramagnetism to ferromagnetism is attributed to an increase of Curie temperature T_c under the tensile strain. When the temperature of measurement was lowered to 160 K (Figure 2b), we observed hysteresis loops exhibited irregularly rectangular shape, which is the indication of labyrinthine-domain ferromagnetic states,^[4] under the strain ranging from 0% to 0.35%. It is noted that the windows of hysteresis loops were enlarged as the strain was increased, showing strain tunable labyrinthine-domain states which could be used to realize strain-assisted magnetization reversal with multiple magnetic states (see Figure S6 in the Supporting Information). Finally, as the temperature was decreased to 130 K, the hysteresis loops exhibited irregularly rectangular shape at zero strain, but gradually transitioned to a rectangular shape as the strain was increased, as shown in Figure 2c. Our results revealed a transition between single- and

labyrinthine-domain states, and thus a shift of the transition temperature T_l which can be controlled by the tensile strain.

We now focus on the strain–temperature phase diagram of the magnetic order in FGT. We first performed the field-cooled (a small external magnetic field $B = 0.03$ T) measurement to extract the strain-dependence of T_c . As shown in Figure 3a, the T_c was determined by the turning points where the remanence of R_{xy} goes to zero.^[1,2] T_c exhibits an increase from 180 to 210 K as the strain is enhanced from 0% to 0.65%, illustrating a significant strain-tunable T_c . In the meanwhile, the transition temperature T_l between labyrinthine- and single-domain FM states can be obtained from the temperature dependent hysteresis loops at different strains. For example, when the tensile strain was set to 0.14%, the hysteresis loop exhibits a rectangular shape below 120 K and becomes irregular above 125 K, as shown in Figure 3b. Therefore, we can take the average temperature, i.e., 122.5 K, as the T_l at the strain of 0.15%, and the temperature interval, which is just 5 K, as the temperature uncertainty. The transitions between labyrinthine- and single-domain states with other strains were shown in Figure S7 (Supporting Information). Based on the strain-dependence of T_c and T_l , we sketched the boundaries between paramagnetic phase, labyrinthine-domain state, and single-domain state, and obtained the phase diagram of FGT flake in the strain–temperature plane, as shown in Figure 3c. It is noted that T_c increases almost linearly with increasing strain, while the transition temperature T_l slowly increases and then saturates as the strain is enhanced. In particular, following the increasing trend of T_c to the tensile strain, we could expect a huge increase of T_c in FGT thin flake at a relatively large strain, since the vdW thin flakes have the capability of undergoing large strains.^[44–46] It is noted that in our experiment, the maximum value of the strain we can achieve is around 0.71%. The main factors that limit the tuning range of the strain are the largely increased contact resistance between the metal electrodes and the FGT sample, as well as the slippage between the FGT sample and PI substrate when a large strain is applied. In order to achieve a larger strain on FGT sample, we may need to replace electrode metals and flexible substrates.

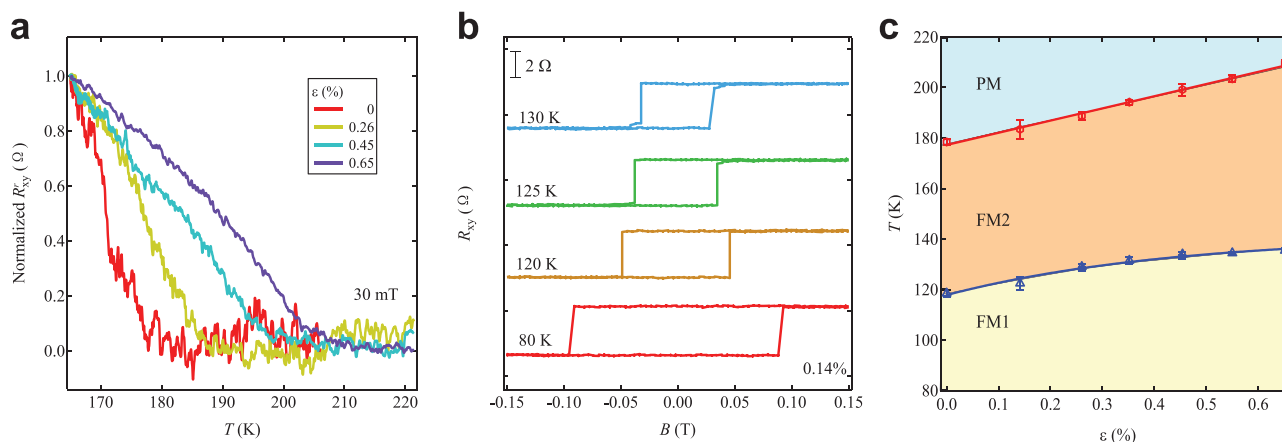


Figure 3. Strain dependence of the Curie temperature (T_c) and magnetic domain states. a) Remanent anomalous Hall resistance R_{xy}^r as a function of temperature with varying strain. R_{xy}^r are normalized by the values at 165 K. b) The R_{xy} versus B curves in the temperature regime from 80 to 160 K with 0.14% strain. c) Phase diagram of FGT with temperature and strain involved. T_c values result from the AHE measurements. PM, FM1, FM2 are labeled as paramagnetic, ferromagnetic with a single-domain, and ferromagnetic with labyrinthine domains. The error bars are the uncertainties in determining the Curie temperatures, T_c , and the transition temperatures between single-domain and labyrinthine domains.

In order to achieve the magnetization reversal, the coercive field H_c has to be reduced below the applied magnetic field by external perturbations, so that the original magnetic state is unable to be maintained due to minimum magnetic energy. The remarkable enhancement of H_c under the strain in FGT provides us a possible way to achieve the strain-assisted magnetization reversals, as conceptually indicated in Figure 4a. We first swept the magnetic field from -0.4 to 0.2 T at 0.71% strain, and the sample was in the magnetic “up” state, as shown in Figure 4b; when the strain was lowered from 0.71% to 0.65%, the sign of R_{xy} flipped from negative to positive, indicating a process of magnetization reversal with the magnetic field kept at 0.2 T. This magnetization reversal process is reversible since we can switch the “up” state back to the “down” state by tuning 0.71–0.65%, as shown in Figure 4c, if the magnetic field was first swept from 0.4 to -0.19 T. Notably, due to the efficient manipulation of coercive force H_c under the tensile strain, the change of strain required for the magnetization reversal in FGT is $\approx 0.06\%$, which is much smaller than that required for the

spin canting in the conventional ferromagnetic layer used in magnetic tunnel junctions.^[47]

Then, we try to investigate the dynamics of the strain change and magnetization reversal in Fe_3GeTe_2 sample, which are characterized by the time-resolved R_{xx} and R_{xy} , respectively. First, we measured the time that R_{xx} and R_{xy} started to change (marked as t_x and t_y in Figure 4b), which indicates the time that the strain begins to change, and the magnetization begins to reverse, respectively. We found that $t_x < t_y$ and a 600 ms delay existed between t_x and t_y . Note that the change of R_{xx} only follows the variation of the strain, and is independent of the magnetization direction (up or down). Thus, we could extract the duration of strain relaxation from the decreasing time of R_{xx} , which is about 6 s as shown in the insets of Figure 4b,c. On the other hand, the duration of magnetization reversal process is ≈ 2.7 s, which is extracted from the time for R_{xy} to flip from down to up (Figure 4b). We also observed an equal duration for the switch from down to up, as shown in Figure 4c, indicating that those values of time delay are reproducible. Notably,

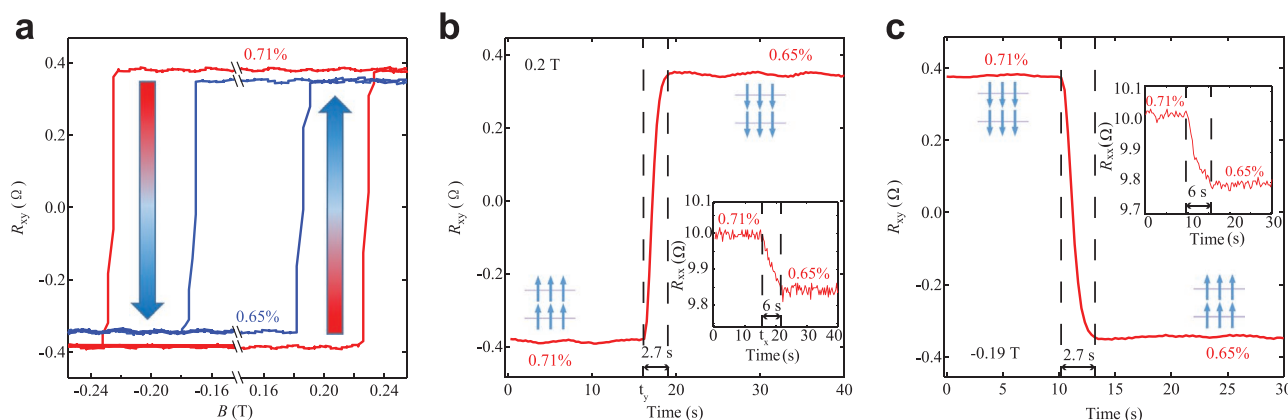


Figure 4. Strain-assisted magnetization reversal. a) The R_{xy} versus B curves with two different strains of 0.65%, 0.71% at 1.5 K. The arrows show the hysteretic loop transformation with the strain changing from 0.71% to 0.65% at 0.2 and -0.19 T. b,c) Magnetization reversal from spin “up” to “down” and “down” to “up” with the strain changing from 0.71% to 0.65% at 0.2 and -0.19 T, respectively. Insets: the time-resolved R_{xx} curve. The t_x and t_y show the time that R_{xx} and R_{xy} started to change. 2.7 and 6 s are the duration time of magnetization reversal and strain relaxation.

our time-resolved data only reflected the local dynamics of the strain-assisted magnetization reversal, while some other interesting topics such as the velocity of domain wall (DW) motion and the mechanism of domain wall nucleation in Fe_3GeTe_2 need more experimental and theoretical studies in the future.

Density functional theory (DFT) calculations were used to elucidate the experimentally observed strain-dependence of magnetic parameters such as H_c and T_c in FGT. Our results suggested that a highly symmetrical FGT bulk structure could undergo transitions with broken structural symmetries when the strain is applied, which releases at least 63 meV per Fe energy (Figure S8c, Supporting Information). The stability of those four low-symmetry phases was also supported by the absence of appreciable imaginary values in calculated vibrational frequencies (see the Experimental Section). Among those symmetry breaking phases, two of them, i.e., the striped and the zigzag phases (Figure S8a,b, Supporting Information), are at least 20 meV per Cr more stable than the undistorted one to the others. The strain-dependent coercive force is usually the most pronounced experimental result that could be directly compared with the calculated strain-dependent magnetic anisotropy energy (MAE), which primarily determines the coercive force^[48,49] and could thus be used to distinguish the exact phase of FGT in our measurement. As shown in Figure 5a, the MAE of the stripe phase slightly enlarges as a function of the applying strain and reaches its maximum at the 0.6% strain, while that of the zigzag phase gently drops (Figure S8d, Supporting Information). After comparing those theoretical results of MAEs with the experimental H_c (see Figure 1e), only the stripe phase reproduces the trend of the experimental data. In light of this, we thus infer the configuration experimentally measured in the single domain phase is, most likely, the stripe phase and thus consider it solely for further discussion.

We then tried to explain our observation that the critical temperatures, i.e., T_c and T_1 , significantly increase as a function of the tensile strain. It has been referred that the MAE in atomically thin FGT flakes plays a key role in maintaining ferromagnetism, and thus significantly influences the T_c by suppressing fluctuations that destroy long-range order.^[1,3] while the spin-exchange couplings (SEC) can be adopted to estimate the Curie temperature in the FGT bulk limit.^[3] We first extracted the SEC

parameters based on a Heisenberg model, which are marked with dashed arrows in different colors in Figure 5b. The intra- and interlayer FM order is the most energetically stable configuration comparing with other magnetic ones (Figure S9, Supporting Information), which is consistent with the experimental observed FM behaviors. Although some SEC parameters indicate AFM couplings, the stronger FM SEC strength and the larger number of FM neighbors dominate, leading to the FM ground state of FGT. Under uniaxial tensile strain along x , those considered intralayer SEC parameters nearly unchanged under different strains and are thus much less influential to the T_c evolutions (Figure 5c). Therefore, the strain-enhanced ferromagnetism in FGT samples is, most likely, not primarily attributed to the change of spin-exchange coupling. In the meanwhile, as discussed above, our calculations show an increase of MAE as the strain is applied within the range from 0.0% to 0.6%, which is consistent with the trend of the increasing Curie temperature in our experimental observation. Therefore, we concluded that the strain-induced enhancement of Curie temperature observed in FGT thin flakes is a result of strain tunable changes of MAE rather than SEC. A previous study of FGT suggested that the MAE is mainly determined by the spin-orbit coupling (SOC),^[21] indicating that the SOC strengthens with respect to the crystal lattice expansion under tensile strain, which is verified by the enlarged orbital portion of the magnetic moment under strain (Figure S10, Supporting Information). It is noted that the T_c continues to rise while the MAE gradually reaches its maximum at 0.6% strain and then reduces under larger strain value, which is most likely, ascribed to the effect of defects and shape anisotropy.^[21,49,50] The magnetism of FGT may originates from both itinerant magnetism and local magnetic moment pictures. In particular, the Stoner FM leads to the magnetism of the Fe^{2+} sublayer and the Fe^{3+} sublayers are governed by the Heisenberg exchange, respectively. As shown in Figure S11 (Supporting Information), we considered the evolution of the Stoner criterion (ST) under uniaxial strain. The strain-dependent trend of ST appears different from the experimental observation of T_c , indicating that the strain-enhanced ferromagnetism is, most likely, not a result of the itinerant magnetism. Uniaxial tensile strain along the y direction leads to similar trends of the MAE and SEC (Figure S12, Supporting Information), which are consistent

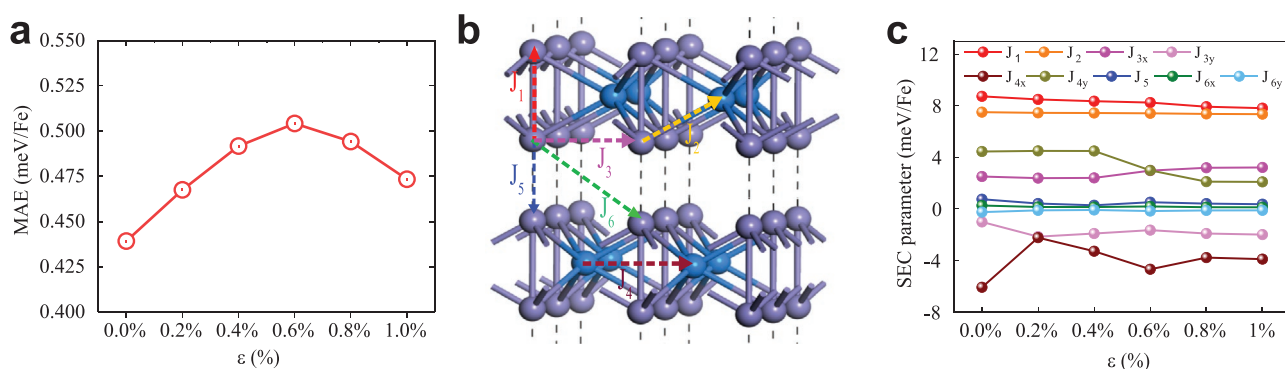


Figure 5. Strain-tuned magnetic properties of FGT from DFT. a) MAE as a function of uniaxial strain along x . b) Perspective view of bulk FGT with SEC parameters between Fe atoms marked with dashed arrows. Only Fe atoms are shown here for better presentation. The slate-blue and blue balls represent the Fe^{3+} and Fe^{2+} , respectively. c) SEC parameters as a function of uniaxial strain along x .

with the experimentally observed similar behaviors for the samples stretched along various orientations.

In summary, we observed the significant enhancement of coercive field in vdW magnet FGT under the strain, and obtained the phase diagram of the magnetic order in the strain–temperature plane. Based on the strong dependence of H_c on strain, we achieved an ultrasensitive strain-assisted magnetization reversal, which may provide an alternative route for the straintronic device. According to the DFT calculations, we attributed those strain-tuning behaviors to the change of MAE. Our strategy of effective strain-tuning on FGT thin flakes can be readily extended to other vdW magnetic systems such as magnetic twist bilayer graphene and layered magnetic topological insulator, etc., offering a platform for studying the physical properties that cannot be accessed by other methods.

Experimental Section

Device Fabrication and Experimental Setup: The flexible substrate used in this research is 200 μm thick polyimide membrane. In the electron-beam lithography process, the conductive adhesive was dropped onto the PMMA layer in order to rule out the electron accumulation. The strain setup was loaded in an Oxford TeslatronPT, which provides magnetic fields up to 12 T and sample temperatures between 1.5 and 300 K. The 4-probe resistance was measured by using a lock-in amplifier (SR830).

DFT Calculations: Density functional theory calculations were performed using the same methods described in our previous publications.^[51–53] In brief, a van der Waals density functional (vdW-DF) method was used,^[54,55] with the optB86b functional^[56] for the exchange part (optB86b-vdW) to optimize atomic structures of the bulk FGT, which usually reveals good agreements of calculated structure-related properties with experimental values of 2D materials.^[57,58] Given optimized structures, the standard Perdew–Burke–Ernzerhof (PBE) functional was used^[59] with consideration of spin–orbit coupling (SOC) to compare energy differences of all considered magnetic configurations; this scheme was found to share the qualitatively same results with the Heyd–Scuseria–Ernzerhof (HSE06) functional^[60] in other magnetic 2D layers, e.g., CrI_3 ^[61] and CrSCL .^[52] Density functional perturbation theory^[62] was employed to calculate vibrational frequencies at the Gamma point (VASP). An on-site Coulomb interaction to the Fe 3d orbitals is considered with different U and J values: $U = 3.6$ eV and $J = 0.2$ eV for Fe^{3+} and $U = 4.6$ eV and $J = 1.0$ eV for Fe^{2+} , as revealed by a linear response method.^[63] In the estimation of Stoner criterion, nonmagnetic calculations were performed to calculate the DOSs using the FM geometries.

Spin Exchange Parameter Estimation: A Heisenberg model was used to model the magnetism of the bulk FGT

$$H = H_0 - \sum_{k=1}^6 J_k \sum_{i,j} \vec{S}_i \cdot \vec{S}_j \quad (1)$$

which includes two in-plane spin-exchange interactions J_1 , J_2 , J_3 , and J_4 , two interlayer ones, J_5 and J_6 (Figure S5b). The spin exchange parameters were derived by the total energy differences of the magnetic configurations with intralayer orders shown in the Figure S12 (Supporting Information) and interlayer FM/AFM orders. Detailed discussion can be found in the caption of Figure S12 (Supporting Information).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

Y.W., C.W., and S.-J.L. contributed equally to this work. The authors thank Junming Liu for useful discussion. This work was supported in part by the National Natural Science Foundation of China (61625402, 61921005, 61974176, 11974156, 11622437, 61674171, and 11974422), the Collaborative Innovation Center of Advanced Microstructures and Natural Science Foundation of Jiangsu Province (BK20190276, BK20180330), Fundamental Research Funds for the Central Universities (020414380122, 020414380084), the Strategic Priority Research Program of Chinese Academy of Sciences (Grant No. XDB30000000), the Fundamental Research Funds for the Central Universities of China and the Research Funds of Renmin University of China (Grants Nos. 16XNLQ01 and 19XNQ025). C.W. was supported by the Outstanding Innovative Talents Cultivation Funded Programs 2017 of Renmin University of China. Calculations were performed at the Physics Lab of High-Performance Computing of Renmin University of China, Shanghai Supercomputer Center. Crystal growth at Rutgers was supported by the center for Quantum Materials Synthesis (cQMS), funded by the Gordon and Betty Moore Foundation's EPIQS initiative through grant GBMF6402, and by Rutgers University.

Conflict of Interest

The authors declare no conflict of interest.

Keywords

coercive field, Fe_3GeTe_2 , magnetization reversal, strain, van der Waals (vdW) magnets

Received: July 3, 2020

Revised: August 8, 2020

Published online: September 11, 2020

- [1] C. Gong, L. Li, Z. Li, H. Ji, A. Stern, Y. Xia, T. Cao, W. Bao, C. Wang, Y. Wang, Z. Q. Qiu, R. J. Cava, S. G. Louie, J. Xia, X. Zhang, *Nature* **2017**, *546*, 265.
- [2] B. Huang, G. Clark, E. Navarro-Moratalla, D. R. Klein, R. Cheng, K. L. Seyler, D. Zhong, E. Schmidgall, M. A. McGuire, D. H. Cobden, W. Yao, D. Xiao, P. Jarillo-Herrero, X. Xu, *Nature* **2017**, *546*, 270.
- [3] Y. Deng, Y. Yu, Y. Song, J. Zhang, N. Z. Wang, Z. Sun, Y. Yi, Y. Z. Wu, S. Wu, J. Zhu, J. Wang, X. H. Chen, Y. Zhang, *Nature* **2018**, *563*, 94.
- [4] Z. Fei, B. Huang, P. Malinowski, W. Wang, T. Song, J. Sanchez, W. Yao, D. Xiao, X. Zhu, A. F. May, W. Wu, D. H. Cobden, J. Chu, X. Xu, *Nat. Mater.* **2018**, *17*, 778.
- [5] M. Bonilla, S. Kolekar, Y. Ma, H. C. Diaz, V. Kalappattil, R. Das, T. Eggers, H. R. Gutierrez, P. Manh-Huong, M. Batzill, *Nat. Nanotechnol.* **2018**, *13*, 289.
- [6] D. R. Klein, D. MacNeill, J. L. Lado, D. Soriano, E. Navarro-Moratalla, K. Watanabe, T. Taniguchi, S. Manni, P. Canfield, J. Fernández-Rossier, P. Jarillo-Herrero, *Science* **2018**, *360*, 1218.
- [7] K. F. Mak, J. Shan, D. C. Ralph, *Nat. Rev. Phys.* **2019**, *1*, 646.
- [8] S. Liang, B. Cheng, X. Cui, F. Miao, *Adv. Mater.* **2019**, *32*, 1903800.
- [9] K. S. Burch, D. Mandrus, J. Park, *Nature* **2018**, *563*, 47.
- [10] Z. Wang, T. Zhang, M. Ding, B. Dong, Y. Li, M. Chen, X. Li, J. Huang, H. Wang, X. Zhao, Y. Li, D. Li, C. Jia, L. Sun, H. Guo, Y. Ye, D. Sun, Y. Chen, T. Yang, J. Zhang, S. Ono, Z. Han, Z. Zhang, *Nat. Nanotechnol.* **2018**, *13*, 554.
- [11] S. Jiang, L. Li, Z. Wang, K. F. Mak, J. Shan, *Nat. Nanotechnol.* **2018**, *13*, 549.

- [12] B. Huang, G. Clark, D. R. Klein, D. MacNeill, E. Navarro-Moratalla, K. L. Seyler, N. Wilson, M. A. McGuire, D. H. Cobden, D. Xiao, W. Yao, P. Jarillo-Herrero, X. Xu, *Nat. Nanotechnol.* **2018**, *13*, 544.
- [13] S. Jiang, J. Shan, K. F. Mak, *Nat. Mater.* **2018**, *17*, 406.
- [14] T. Song, Z. Fei, M. Yankowitz, Z. Lin, Q. Jiang, K. Hwangbo, Q. Zhang, B. Sun, T. Taniguchi, K. Watanabe, M. A. McGuire, D. Graf, T. Cao, J. Chu, D. H. Cobden, C. R. Dean, D. Xiao, X. Xu, *Nat. Mater.* **2019**, *18*, 1298.
- [15] T. Li, S. Jiang, N. Sivadas, Z. Wang, Y. Xu, D. Weber, J. E. Goldberger, K. Watanabe, T. Taniguchi, C. J. Fennie, K. Fai Mak, J. Shan, *Nat. Mater.* **2019**, *18*, 1303.
- [16] Z. Li, M. Zhang, T. Hou, J. Zhao, L. Li, A. Rahman, Z. Xu, J. Gong, Z. Chi, R. Dai, Z. Wang, Z. Qiao, Z. Zhang, X. Wang, *Phys. Rev. B* **2019**, *100*, 14407.
- [17] M. Gibertini, M. Koperski, A. F. Morpurgo, K. S. Novoselov, *Nat. Nanotechnol.* **2019**, *14*, 408.
- [18] P. Sheng, B. Wang, R. Li, *J. Semicond.* **2018**, *39*, 011006.
- [19] A. A. Bukharaev, A. K. Zvezdin, A. P. Pyatakov, Y. K. Fetisov, *Phys.-Usp.* **2018**, *61*, 1175.
- [20] N. D. Souza, A. Biswas, H. Ahmad, M. S. Fashami, M. M. Al-Rashid, V. Sampath, D. Bhattacharya, M. A. Abeed, J. Atulasimha, S. Bandyopadhyay, *Nanotechnology* **2018**, *29*, 065604.
- [21] H. L. Zhuang, P. R. C. Kent, R. G. Hennig, *Phys. Rev. B* **2016**, *93*, 134407.
- [22] Z. Wu, J. Yu, S. Yuan, *Phys. Chem. Chem. Phys.* **2019**, *21*, 7750.
- [23] J. Yan, L. Webster, *Phys. Rev. B* **2018**, *98*, 144411.
- [24] X. Li, J. Yang, *J. Mater. Chem. C* **2014**, *2*, 7071.
- [25] L. Mayer, *J. Appl. Phys.* **1958**, *29*, 1003.
- [26] D. Chiba, M. Yamanouchi, F. Matsukura, H. Ohno, *Science* **2003**, *301*, 943.
- [27] D. B. Gopman, C. L. Dennis, P. J. Chen, Y. L. Iunin, P. Finkel, M. Staruch, R. D. Shull, *Sci. Rep.* **2016**, *6*, 27774.
- [28] K. Kim, J. Seo, E. Lee, K. T. Ko, B. S. Kim, B. G. Jang, J. M. Ok, J. Lee, Y. J. Jo, W. Kang, J. H. Shim, C. Kim, H. W. Yeom, B. Il Min, B. Yang, J. S. Kim, *Nat. Mater.* **2018**, *17*, 794.
- [29] N. León-Brito, E. D. Bauer, F. Ronning, J. D. Thompson, R. Movshovich, *J. Appl. Phys.* **2016**, *120*, 083903.
- [30] J. Zhu, M. Janoschek, D. S. Chaves, J. C. Cezar, T. Durakiewicz, F. Ronning, Y. Sassa, M. Mansson, B. L. Scott, N. Wakeham, E. D. Bauer, J. D. Thompson, *Phys. Rev. B* **2016**, *93*, 144404.
- [31] B. Chen, J. Yang, H. Wang, M. Imai, H. Ohta, C. Michioka, K. Yoshimura, M. Fang, *J. Phys. Soc. Jpn.* **2013**, *82*, 124711.
- [32] G. D. Nguyen, J. Lee, T. Berlijn, Q. Zou, S. M. Hus, J. Park, Z. Gai, C. Lee, A. Li, *Phys. Rev. B* **2018**, *97*, 014425.
- [33] Y. Wang, C. Xian, J. Wang, B. Liu, L. Ling, L. Zhang, L. Cao, Z. Qu, Y. Xiong, *Phys. Rev. B* **2017**, *96*, 134428.
- [34] K. S. Burch, D. Mandrus, J. Park, *Nature* **2018**, *563*, 47.
- [35] S. Liu, X. Yuan, Y. Zou, Y. Sheng, C. Huang, E. Zhang, J. Ling, Y. Liu, W. Wang, C. Zhang, J. Zou, K. Wang, F. Xiu, *npj 2D Mater* **2017**, *1*, 30.
- [36] C. Tan, J. Lee, S. Jung, T. Park, S. Albarakati, J. Partridge, M. R. Field, D. G. McCulloch, L. Wang, C. Lee, *Nat. Commun.* **2018**, *9*, 1554.
- [37] J. H. Kang, C. Park, J. A. Scholl, A. H. Brazin, N. M. Holloway, J. W. High, S. E. Lowther, J. S. Harrison, *J. Polym. Sci., Part B: Polym. Phys.* **2009**, *47*, 994.
- [38] Y. Wang, A. X. Wang, Y. Wang, M. K. Chyu, Q. Wang, *Sensor. Actuators, A* **2013**, *199*, 265.
- [39] R. Roldán, A. Castellanos-Gomez, E. Cappelluti, F. Guinea, *J. Phys.: Condens. Matter* **2015**, *27*, 313201.
- [40] N. Petrone, I. Meric, J. Hone, K. L. Shepard, *Nano Lett.* **2013**, *13*, 121.
- [41] H. J. Conley, B. Wang, J. I. Ziegler, R. F. Haglund, S. T. Pantelides, K. I. Bolotin, *Nano Lett.* **2013**, *13*, 3626.
- [42] Z. Zhang, L. Li, J. Horng, N. Z. Wang, F. Yang, Y. Yu, Y. Zhang, G. Chen, K. Watanabe, T. Taniguchi, X. H. Chen, F. Wang, Y. Zhang, *Nano Lett.* **2017**, *17*, 6097.
- [43] Y. Zhang, L. Shen, M. Liu, X. Li, X. Lu, L. Lu, C. Ma, C. You, A. Chen, C. Huang, L. Chen, M. Alexe, C. Jia, *ACS Nano* **2017**, *11*, 8002.
- [44] S. Deng, A. V. Sumant, V. Berry, *Nano Today* **2018**, *22*, 14.
- [45] C. Lee, X. Wei, Q. Li, R. Carpick, J. W. Kysar, J. Hone, *Phys. Status Solidi B* **2009**, *246*, 2562.
- [46] J. Annett, G. L. W. Cross, *Nature* **2016**, *535*, 271.
- [47] P. Li, A. Chen, D. Li, Y. Zhao, S. Zhang, L. Yang, Y. Liu, M. Zhu, H. Zhang, X. Han, *Adv. Mater.* **2014**, *26*, 4320.
- [48] E. C. Stoner, E. P. Wohlfarth, *Philos. Trans. R. Soc., A* **1948**, *240*, 599.
- [49] C. Tannous, J. Gieraltowski, *Eur. J. Phys.* **2008**, *29*, 475.
- [50] M. T. Johnson, P. J. H. Bloemen, F. J. A. D. Broeder, J. J. D. Vries, *Rep. Prog. Phys.* **1996**, *59*, 1409.
- [51] X. Zhou, Y. Pan, J. Qiao, X. Kong, C. Kaun, W. Ji, C. Wang, *Phys. Rev. B* **2018**, *97*, 245409.
- [52] C. Wang, X. Zhou, L. Zhou, N. Tong, Z. Lu, W. Ji, *Sci. Bull.* **2019**, *64*, 293.
- [53] C. Wang, D. Chen, Z. Zhong, Z. Yuan, Z. Lu, W. Ji, P. Jiang, *Phys. Rev. B* **2019**, *99*, 144401.
- [54] H. Rydberg, E. Schröder, D. C. Langreth, B. I. Lundqvist, M. Dion, *Phys. Rev. Lett.* **2004**, *92*, 246401.
- [55] É. D. Murray, L. Kong, B. I. Lundqvist, D. C. Langreth, K. Lee, *Phys. Rev. B* **2010**, *82*, 81101.
- [56] D. R. Bowler, A. Michaelides, J. Klimeš, *Phys. Rev. B* **2011**, *83*, 195131.
- [57] J. Qiao, Y. Pan, F. Yang, C. Wang, Y. Chai, W. Ji, *Sci. Bull.* **2018**, *63*, 159.
- [58] Z. Hu, X. Kong, J. Qiao, B. Normand, W. Ji, *Nanoscale* **2016**, *8*, 2740.
- [59] K. Burke, M. Ernzerhof, J. P. Perdew, *Phys. Rev. Lett.* **1996**, *77*, 3865.
- [60] J. Heyd, G. E. Scuseria, M. Ernzerhof, *J. Chem. Phys.* **2003**, *118*, 8207.
- [61] P. Jiang, C. Wang, D. Chen, Z. Zhong, Z. Yuan, Z. Lu, W. Ji, *Phys. Rev. B* **2019**, *99*, 144401.
- [62] S. de Gironcoli, A. Dal Corso, P. Giannozzi, S. Baroni, *Rev. Mod. Phys.* **2001**, *73*, 515.
- [63] S. de Gironcoli, M. Cococcioni, *Phys. Rev. B* **2005**, *71*, 35105.